

# BASICS FOR AIR TRAFFIC CONTROL – NOTICES TO AIRMEN (NOTAMs)

## INTRODUCTION

Conditions of airports, Navigational Aids (NAVAIDs), Air Traffic Control (ATC) facilities, and services to the aviation community are always changing. When the changes occur on a scheduled basis, they are published in the Aeronautical Information System. However, not all changes occur in a timely manner. When this happens, the Notice to Airmen (NOTAM) System is used to notify the aviation community.

This module discusses how time-critical information concerning the National Airspace System (NAS) is received, processed, and disseminated to all aviation interests.



## NOTICES TO AIRMEN (NOTAMs)

**Purpose:** This lesson discusses how time-critical information concerning the National Airspace System is received, processed, and disseminated to all aviation interests.

### Objectives:

- Identify methods of disseminating airmen's information
- Identify types of NOTAMs
- Describe responsibilities for reporting different classifications of NOTAMs

References for this lesson are as follows:

- FAA Order JO 7110.65, Air Traffic Control
- FAA Order JO 7930.2, Notices to Airmen (NOTAM)

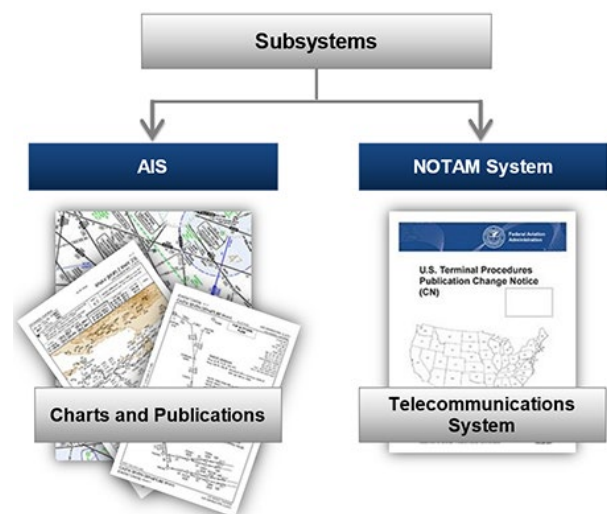
### Dissemination of Airmen's Information

The system for disseminating aeronautical information is made up of two subsystems:

- The Aeronautical Information System (AIS), which consists of charts and publications
- The NOTAM system, which is a telecommunication system

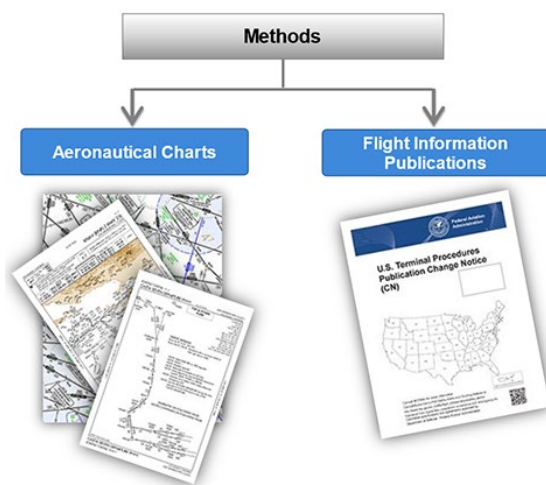
*Charts & Publications are updated every 56 days.*

*NOTAMs are used in the interim*



Airmen's information is disseminated by the following methods:

- Aeronautical Charts
- Flight Information Publications in book form



| AERONAUTICAL CHARTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | FLIGHT INFORMATION PUBLICATIONS                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>■ Instrument Flight Rules (IFR)<ul style="list-style-type: none"><li>• En Route High Altitude Charts</li><li>• En Route Low Altitude Charts</li></ul></li><li>■ Visual Flight Rules (VFR)<ul style="list-style-type: none"><li>• Sectional Aeronautical Charts (SACs)</li><li>• Terminal Area Charts (TACs)</li></ul></li><li>■ U.S. Terminal Procedures Publication (TPP)<ul style="list-style-type: none"><li>• Departure Procedures (DP)</li><li>• Standard Terminal Arrivals (STAR)</li><li>• Standard Instrument Approach Procedures (SIAPs)</li></ul></li></ul> | <ul style="list-style-type: none"><li>■ Chart supplements</li></ul> |

## National Flight Data Center (NFDC)

The National Flight Data Center (NFDC) is located in Washington, DC. The NFDC was established by the FAA to operate a central aeronautical information service for the collection, validation, and dissemination of aeronautical data that supports the government, industry, and the aviation community.



Originators of airmen's information are expected to inform the NFDC in sufficient time before the effective dates of changes to permit publishing of aeronautical data on the various charts or in the appropriate publications.



## Knowledge Check A

REVIEW what you have learned so far about dissemination of airmen's information. ANSWER the question below.

1. What are the two subsystems for disseminating aeronautical information? (Select the correct answer.)

- ☐ Aeronautical Charts and AIS
- ☐ NOTAM System and Flight Information Publication
- ☒ AIS and NOTAM System

## Notice to Airmen

A NOTAM is a notice containing information (not known sufficiently in advance to publicize by other means) concerning the establishment, condition, or change in any component (facility, service, or procedure of, or hazard in the NAS), the timely knowledge of which is essential to personnel concerned with flight operations.

KATL HARTSFIELD - JACKSON ATLANTA INTL

IATL 05/241 (KATL A1876/17) ATL RWY 28 FICON 5/5/5 100 PCT WET OBS AT 1705231518. 1705231624-1705241624

IATL 05/236 (KATL A1872/17) ATL NAV TACAN OUT OF SERVICE 1705251300-1705251700

IATL 05/214 (KATL A1837/17) ATL APRON 3 NORTHEAST RAMP CLSD NORTH OF GATE C50 1705220330-1706071030

IATL 05/192 ATL OBST TOWER LGT (ASR 1019346) 333842.50N0841110.40W (12.1NM E ATL) 1018.7FT (261.8FT AGL) OUT OF SERVICE 1705191639-1706191639

IATL 05/176 (KATL A1794/17) ATL TWY F3 CLSD 1705181945-1706031000

IATL 04/174 ATL AIRSPACE CONTROLLED BURN WI AN AREA DEFINED AS .5NM RADIUS ATL135000.9 SFC-2000FT AGL THU 1200-1600 1704271200-1706291600

IATL 04/107 ATL COM REMOTE TRANSMITTER/RECEIVER 239.275 CHANGED TO 257.9 1604191648-PERM

IFDC 7/7053 (KATL A1528/17) ATL IAP HARTSFIELD - JACKSON ATLANTA INTL, Atlanta, GA. ILS OR LOC RWY 8R, AMDT 61... ILS PRM RWY 8R (SIMULTANEOUS CLOSE PARALLEL), AMDT 2... MISSED APPROACH: CLIMB TO 1500 THEN CLIMBING LEFT TURN TO 4000 DIRECT PDK VOR/DME AND HOLD, ATL VOR OUT OF SERVICE. 1704280534-1712080534EST

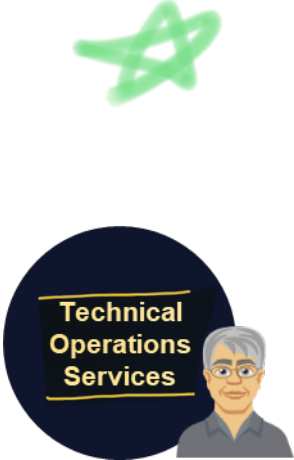
All air traffic employees, regardless of position, must immediately report any situation or condition considered hazardous to flight to an air traffic facility for appropriate action.

A **Certified Source** is the party who enters/submits a NOTAM to the United States NOTAM System (NS), using an approved direct entry tool or interface.

The Certified Source is responsible for the correct classification and format of the NOTAM and for ensuring that facilities affected by the NOTAM are aware of the new NOTAM.

## Responsibilities

Certified Sources

|                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <p>Air traffic personnel must accept all airmen information regardless of source or subject matter, provided the occurrence is no more than 3 days in the future.</p> <ul style="list-style-type: none"> <li>Obtain the name, title (if appropriate), address, and telephone number of the person furnishing the information and forward all data to the appropriate Flight Service Station (FSS) for NOTAM issuance</li> </ul> <p>FSSs, Terminals, and Air Route Traffic Control Centers (ARTCCs) accept and disseminate NOTAMs.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|   | <p>Technical Operations Services, Operations Center manager, or representative, is responsible for:</p> <ul style="list-style-type: none"> <li>Originating and canceling NOTAM information for shutdown, restoration, or any condition that affects the operations of navigational aids (NAVAIDs), frequencies, or other electronic aids that affect safety of flight</li> </ul> <p>Known or reported malfunctions of a NAVAID must be reported to Technical Operations or appropriate personnel.</p> <p>Information concerning maintenance shutdown of NAVAIDs that are part of the NAS must be handled as follows:</p> <ul style="list-style-type: none"> <li><b>Routine maintenance shutdown</b> – When possible, approval should be obtained sufficiently in advance of the proposed shutdown time to allow dissemination of a NOTAM <b>at least 5 hours before a shutdown</b> will occur (See Note)</li> <li><b>Emergency shutdown</b> – When possible, obtain <b>at least 1 hour advance notice</b> so that appropriate dissemination may be made before shutdown</li> <li><b>Extended maintenance shutdown</b> – Notify the NFDC sufficiently in advance to permit publication of the information prior to the shutdown date</li> </ul> <p><b>Note:</b> Approval is obtained from the ATC facility affected by the maintenance shutdown.</p> |
|  | <p>Airport management is responsible for observing and reporting the condition of the aerodrome services, facilities, and movement area.</p> <ul style="list-style-type: none"> <li>The FSS air traffic managers (ATMs) must coordinate with appropriate airport managers to obtain a list of airport employees who are authorized to originate NOTAMs</li> </ul> <p>Only airport management can close any portion of an airport.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



## Authority

- The United States NOTAM Office (USNOF) is the authority ensuring NOTAM formats and is charged with monitoring the NOTAM System (NS)
- All NOTAMs will be processed, stored, and distributed by the NS
- The USNOF may edit any NOTAM (except FDC NOTAMs); the contents of a NOTAM must not be changed without notifying the originating facility
- NOTAMs must have one of the following keywords as the first part of the text; a keyword is used to make it easier to sort and locate the specific data needed

### KATL HARTSFIELD - JACKSON ATLANTA INTL

IATL 05/241 (KATL A1876/17) ATL RWY 28 FICON 5/5/5 100 PCT WET OBS AT 1705231518. 1705231624-1705241624

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RWY – Runway

TWY – Taxiway

APRON – Apron *Non-Movement Area*

AD – Aerodrome

OBST – Obstructions

NAV – Navigation

COM – Communications

SVC – Services

AIRSPACE – Airspace

ODP – Obstacle Departure Procedure

SID – Standard Instrument Departure

STAR – Standard Terminal Arrival

CHART – Chart

DATA – Data

IAP – Instrument Approach Procedure

VFR – Visual Flight Procedure

ROUTE – Route

SPECIAL – Special

SECURITY – Security

## Unverified NOTAMs

Unverified aeronautical information is used only when authorized by Letters of Agreement (LOAs). An unverified NOTAM will have a (U) in it. A movement area or other information received that meets NOTAM criteria but has not been confirmed by the airport manager or designee would be unverified.

!TEX 06/013 TEX (U) RWY 9/27 UNSAFE DISABLED ACFT W END 1706302124-1707010600



## Knowledge Check B

REVIEW what you have learned so far about Notices to Airmen. ANSWER the questions below.

1. Which subsystem is used for disseminating aeronautical information? (Select the correct answer.)
  - ☒ The AIS
  - ☐ Jeppesen Database
  - ☐ Technical Operations Services
2. Who submits a NOTAM to the NS using an approved direct entry tool or interface? (Select the correct answer.)
  - ☐ NOTAM Office
  - ☐ Airport Management
  - ☒ Certified Source
3. Who is responsible for reporting a condition considered hazardous to flight? (Select the correct answer.)
  - ☒ All air traffic personnel
  - ☐ Technical Operations Services
  - ☐ Airport Management

## NOTAM Classification



NOTAMs are classified into the following groups:

- NOTAM D
- FDC NOTAM
- Pointer NOTAM
- FICON NOTAM

## NOTAM D

Wide



A **NOTAM (D)** is a classification of NOTAMs containing information concerning the establishment, condition, or change in any aeronautical facility, en route navigational aids, services, procedures, hazards and civil public-use airports listed in the Chart Supplement.

- NOTAM Ds are issued when the information requires wide dissemination via telecommunication and pertains to en route NAVAIDS, civil public-use airports listed in the chart supplements, facilities, services, and procedures
- Notice distributed by means of telecommunications, containing information concerning the establishment, condition, or change in any aeronautical facility, service, procedure, or hazard, the timely knowledge of which is essential to personnel concerned with flight operations

### EXAMPLES

!LMT 04/053 LMT TWY D BTN TWY E AND TWY F CLSD 1604271936-PERM

!TYR 02/017 TYR RWY 04 RWY END ID LGT OUT OF SERVICE 1702241244-1803311700

!SAN 05/060 JLI OBST MOORED BALLOON WI AN AREA DEFINED AS 1NM RADIUS OF JLI263021 (5.4NM NW RNM) 980FT (500FT AGL) 1605261629-1705261620

## Flight Data Center (FDC) NOTAM



**FDC NOTAM** – The classification of NOTAMs containing flight information that is normally regulatory in nature including, but not limited to, changes to IFR charts, procedures, and airspace usage.

FDC NOTAMs are issued for:

- Interim IFR flight procedures

Air traffic service route changes

Instrument flight procedure changes

Airspace changes in general

- Temporary flight restrictions

Disaster/hazard areas

Aerial demonstrations

Hijacking

- Flight restrictions in the proximity of the President and other parties

The NOTAM-issuing authority is responsible for canceling FDC NOTAMs

- An FDC NOTAM must be issued to cancel an FDC NOTAM

### EXAMPLES

**!FDC 7/6030** SJC ODP NORMAN Y MINETA SAN JOSE INTL, SAN JOSE, CA. TAKEOFF MINIMUMS AND (OBSTACLE) DEPARTURE PROCEDURES AMDT 6C... TAKEOFF MINIMUMS: RWY 12L/R, 400-2 1/2 OR STANDARD WITH A MINIMUM CLIMB OF 315FT PER NM TO 500. ADD NOTE: RWY 12L, TEMPORARY CRANE 1.28NM FROM DER, 722FT LEFT OF CENTERLINE, 284FT AGL/ 368FT MSL (2016-AWP-1614-OE). RWY 12R, TEMPORARY CRANE 1.28NM FROM DER, 1472FT LEFT OF CENTERLINE, 284FT AGL/ 368FT MSL (2016-AWP-1614-OE). ALL OTHER DATA REMAINS AS PUBLISHED. 1703211455-1704011455EST

## Pointer NOTAM



**Pointer NOTAMs** are NOTAM D issued to point to additional aeronautical information. The keyword in the pointer NOTAM must match the keyword in the original NOTAM.

Pointer NOTAMs are issued by a flight service station to highlight or point out another NOTAM; such as an FDC or Parachute Jump Exercise (PJE) NOTAM.

- This type of NOTAM will assist users in cross-referencing important information that may NOT be found under an airport or NAVAID identifier

- Keywords in pointer NOTAMs must match the keywords in the NOTAM D that is being pointed out

- Keywords in pointer NOTAMs related to temporary flight restrictions (TFR) must be AIRSPACE

### EXAMPLES

**Main NOTAM** – !MKL 07/195 GHM NAV VORTAC DECOMMISSIONED 1507171856-PERM

**Pointer NOTAM** – !MKL 02/164 MRC NAV SEE MKL 07/195 VORTAC DECOMMISSIONED 1602191405-PERM

## Field Condition (FICON) NOTAM



FICON NOTAMs are used to report the surface conditions on runways, taxiways, and aprons.

- They are also used to report surface conditions of a helipad

### EXAMPLES

AD HELIPORT **FICON 4IN DRY SN** OBSERVED AT XX12051621

TWY A **FICON 1/2IN WET SN**

APRON MAIN RAMP BTN RWY 13/31 ABD TWY C **FICON 1IN DRY SN**



### Knowledge Check C

*REVIEW what you have learned so far about NOTAM classification and Notice to Airmen publications. ANSWER the questions below.*

1. Why are NOTAM Ds issued? *(Select the correct answer.)*
  - ☐ To cancel an FDC NOTAM
  - ☒ To disseminate information over a wide area
  - ☐ To highlight or point out another NOTAM
2. What NOTAMs are used to report the surface conditions on runways, taxiways, and aprons? *(Select the correct answer.)*
  - ☐ Unverified NOTAMs
  - ☒ FICON NOTAMs
  - ☐ NOTAM D
3. What type of NOTAM is issued for aerial demonstrations? *(Select the correct answer.)*
  - ☐ NOTAM D
  - ☐ Pointer NOTAM
  - ☒ FDC NOTAM



## SUMMARY

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This lesson discussed how time-critical information concerning the National Airspace System is received, processed, and disseminated to all aviation interests.

In accordance with FAA Order JO 7110.65, Air Traffic Control and FAA Order JO 7930.2, Notices to Airmen (NOTAM), you should now be able to:

- Identify methods of disseminating airmen's information
- Identify types of NOTAMs
- Describe responsibilities for reporting different classifications of NOTAMs

## KNOWLEDGE CHECK ANSWER KEY

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### Knowledge Check A

1. AIS and NOTAM System

### Knowledge Check B

1. The AIS; 2. Certified Source; 3. All air traffic personnel

### Knowledge Check C

1. To disseminate information over a wide area; 2. FICON NOTAMs; 3. FDC NOTAM